2025 VCE Specialist Mathematics Year 12 Trial Examination 1

Detailed Answers

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Question 1

$\cos(2x) = 2\cos^2(x) - 1 = |\cos(x)|$

Consider when

$\cos(x) \geq 0, \quad |\cos(x)| = \cos(x)$

$\cos(2x) = 2\cos^2(x) - 1 = \cos(x)$

$2\cos^2(x) - \cos(x) - 1 = 0$

$(2\cos(x)+1)(\cos(x)-1) = 0$

$\cos(x) = 1, \quad \cos(x) \geq 0$

$x = 2n\pi, \quad n \in Z$

Consider when

$\cos(x) \leq 0 \quad |\cos(x)| = -\cos(x)$

$\cos(2x) = 2\cos^2(x) - 1 = -\cos(x)$

$2\cos^2(x) + \cos(x) - 1 = 0$ M2

$(2\cos(x)-1)(\cos(x)+1) = 0$

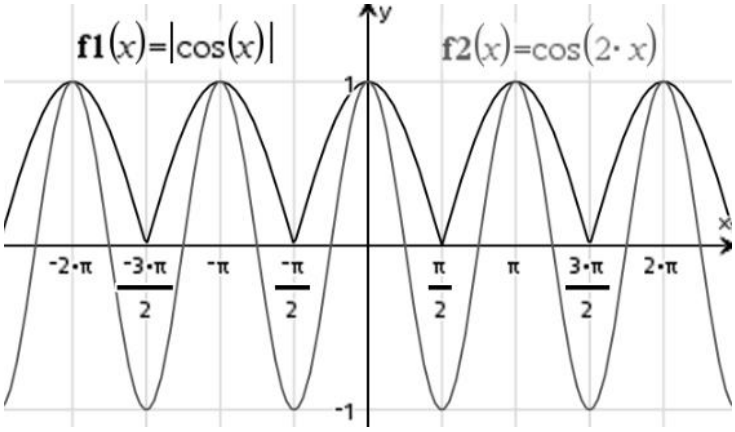
$\cos(x) = -1, \quad \cos(x) \leq 0$

$x = (2n+1)\pi, \quad n \in Z$

or graphically

$\cos(2x) = |\cos(x)|$

$x = n\pi \quad n \in Z$ A1



Question 2

To prove $\log_2(3)$ is irrational, we use a proof by contradiction.

Assume that $\log_2(3)$ is rational, since $\log_2(3) > 1$, we can write M1

$\log_2(3) = \dfrac{a}{b}, \quad a, b \in Z^+$

$2^{\frac{a}{b}} = 3$ from the definition of the log

$2^a = 3^b$ raising both sides to the bth power

but the left hand side 2^a where $a \in Z^+$ is always an even number, whereas

the right hand side 3^b where $b \in Z^+$ is always an odd number, A1

hence we have a contradiction.

So our original assumption that is that $\log_2(3)$ is rational must be false,

Hence $\log_2(3)$ is irrational.

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Question 3

a. $f: R\setminus\{1\} \to R,\ \ f(x) = \dfrac{x^2-4}{x-1}$

$f(x) = \dfrac{x^2-4}{x-1} = \dfrac{x^2-x+x-1-3}{x-1} = \dfrac{x(x-1)+x-1-3}{x-1} = x+1-\dfrac{3}{x-1}$,

so $y = x+1$ is an oblique asymptote and $x = 1$ is a vertical asymptote. A1

b. $f(x) = x+1-3(x-1)^{-1}$

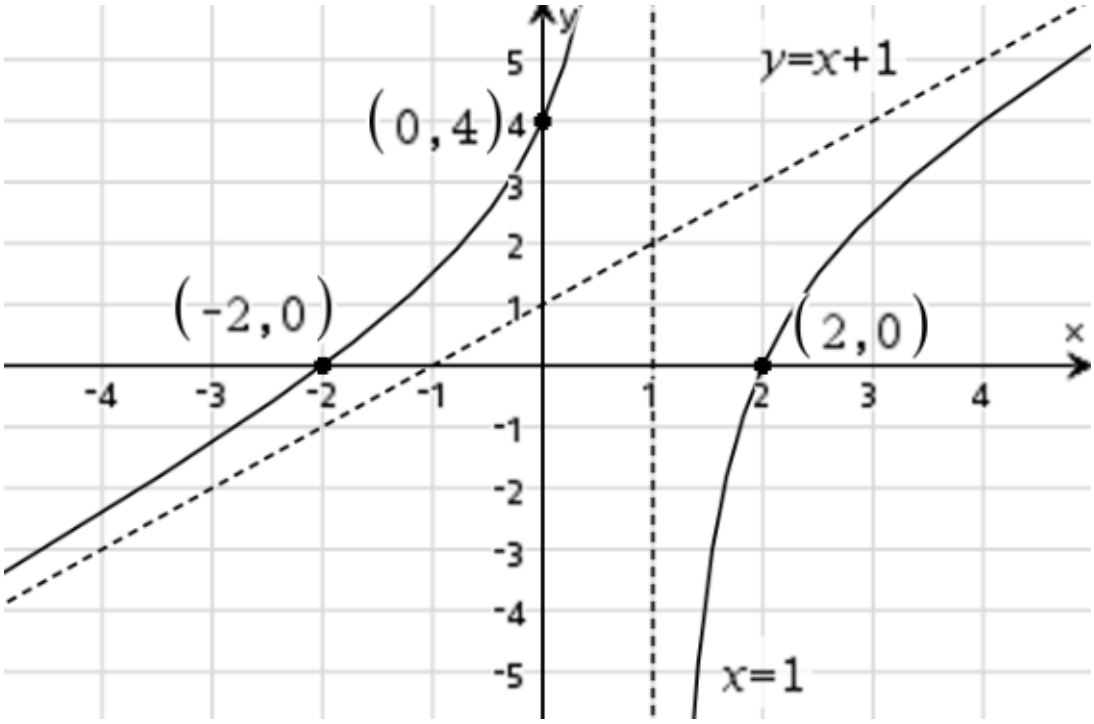
$f'(x) = 1+3(x-1)^{-2} = 1+\dfrac{3}{(x-1)^2} \neq 0$, so there are no turning points.

$f''(x) = -6(x-1)^{-3} = \dfrac{-6}{(x-1)^3} \neq 0$, so there are no points of inflection. A1

c. the graph crosses the x-axis when $x^2-4=0,\quad x=\pm 2,\quad (-2,0),(2,0)$

the graph crosses the y-axis $f(0) = \dfrac{-4}{-1} = 4,\quad (0.4)$

correct graph, shape, asymptotes, axial intercepts G2



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Question 4

a. When $x=-1,\quad t=2,\quad v=\dfrac{1}{-(4+4)}=-\dfrac{1}{8}$, find the acceleration $a=\dfrac{d^2x}{dt^2}$

$x\dfrac{dx}{dt}=\left(4+t^2\right)^{-1}$ using implicit differentiation, differentiating with respect to t.

$$\frac{d}{dt}\left(x\frac{dx}{dt}\right)=\frac{d}{dt}\left(\left(4+t^2\right)^{-1}\right)$$

$$\left(\frac{dx}{dt}\right)^2+x\frac{d^2x}{dt^2}=\frac{-2t}{\left(4+t^2\right)^2}$$ M1

$v^2+xa=\dfrac{-2t}{\left(4+t^2\right)^2}$ substitute $x=-1,\quad t=2,\quad v=-\dfrac{1}{8}$ A1

$$\frac{1}{64}-a=\frac{-4}{(4+4)^2}=\frac{-1}{16}$$

$$a=\frac{1}{64}+\frac{1}{16}$$

$$a=\frac{5}{64}\ \text{m s}^{-2}$$ A1

b. $v=\dfrac{dx}{dt}=\dfrac{1}{x\left(4+t^2\right)},\quad x(2)=-1$

$$\int x\,dx=\int\frac{1}{\left(4+t^2\right)}dt$$ M1

$$\frac{1}{2}x^2=\frac{1}{2}\tan^{-1}\left(\frac{t}{2}\right)+c$$

$x=-1,\quad t=2\quad\Rightarrow\quad\dfrac{1}{2}=\dfrac{1}{2}\tan^{-1}(1)+c$ A1

$$c=\frac{1}{2}-\frac{\pi}{8}$$

$$x^2=\tan^{-1}\left(\frac{t}{2}\right)+1-\frac{\pi}{4}$$

$$x=\pm\sqrt{\tan^{-1}\left(\frac{t}{2}\right)+1-\frac{\pi}{4}}$$

but $x<0$, when $t=2$, so

$$x=-\sqrt{\tan^{-1}\left(\frac{t}{2}\right)+1-\frac{\pi}{4}}$$ A1

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Question 5

$x = 2t^3,\ \ y = 3t^2,$ for $0 \le t \le 2$

$$S_x = 2\pi \int_{t_1}^{t_2} y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$\frac{dx}{dt} = 6t^2,\ \ \frac{dy}{dt} = 6t$$

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 36t^4 + 36t^2 = 36t^2\left(t^2+1\right),$$ M1

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = 6|t|\sqrt{t^2+1},\quad 0 \le t \le 2$$

So the modulus is not needed.

$$S_x = 2\pi\int_0^2 3t^2 \times 6t\sqrt{1+t^2}\, dt$$ A1

$$S_x = 36\pi\int_0^2 t \times t^2\sqrt{1+t^2}\, dt$$

let $u = 1+t^2\ \ \dfrac{du}{dt} = 2t,\quad t^2 = u-1$ M1

terminals $t = 0,\ u = 1,\quad t = 2,\ u = 5$

$$S_x = 18\pi\int_1^5 (u-1)\sqrt{u}\, du,\qquad k = 18,\quad a = 1,\quad b = 5,\quad c = 1$$ A1

Question 6

a. steak $S \overset{d}{=} N\left(350, 4^2\right)$ bone $B \overset{d}{=} N\left(60, 2.5^2\right)$ fat $F \overset{d}{=} N\left(20, 1.5^2\right)$

edible part $T = S - B - F$

$$\mathrm{E}(T) = \mathrm{E}(S) - \mathrm{E}(B) - \mathrm{E}(F) = 350 - 60 - 20 = 270$$

$$\mathrm{Var}(T) = \mathrm{Var}(S) + \mathrm{Var}(B) + \mathrm{Var}(F) = 16 + \left(\frac{5}{2}\right)^2 + \left(\frac{3}{2}\right)^2 = \frac{64+25+9}{4} = \frac{98}{4}$$

$2T = T_1 + T_2,\qquad \mathrm{E}(2T) = 2\mathrm{E}(T) = 2 \times 270 = 540$ A1

$\mathrm{Var}(2T) = 2\mathrm{Var}(T) = 2 \times \dfrac{98}{4} = \dfrac{98}{2} = 49,\ \ \sigma_{2T} = 7,\quad 2T \overset{d}{=} N(540, 49)$ A1

b. $\Pr(2T > 526) = \Pr\left(Z > \dfrac{526-540}{7}\right)$

$$= \Pr(Z > -2) = \Pr(Z < 2) = \Pr(-2 < Z < 2) + \Pr(Z > 2)$$

$$= 0.95 + \frac{1}{2}(1 - 0.95)$$

$$= 0.975$$ A1

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Question 7

To prove $3^{2n+1}+2^{n+2}$ is divisible by 7 for all $n \in N$.

Base case $n=1$ $LHS = 3^3+2^3 = 27+8 = 35 = 5\times 7$. This is divisible by 7. A1

The statement is true when $n=1$

Assume it is true when $n=k$ so that $3^{2k+1}+2^{k+2}$ is divisible by 7, so that

we can write $3^{2k+1}+2^{k+2}=7a$ where $a \in N$, M1

Now consider

$3^{2(k+1)+1}+2^{(k+1)+2}$

$=3^{2k+3}+2^{k+3}$

$=3^{2k+1+2}+2^{k+2+1}$

$=3^2\times 3^{2k+1}+2^1\times 2^{k+2}$ M1

$=9\times 3^{2k+1}+2\times 2^{k+2}$

$=7\times 3^{2k+1}+2\times 3^{2k+1}+2\times 2^{k+2}$

$=7\times 3^{2k+1}+2\times\left(3^{2k+1}+2^{k+2}\right)$

$=7\times 3^{2k+1}+2\times 7a$

$=7\left(3^{2k+1}+2a\right)$

$=7b$ $b \in N$, so it is divisible by 7. A1

Since it is true when $n=1$ and assuming it is true when $n=k$ then it is true when $n=k+1$, so by the principle of mathematical induction, it is true for $n \in N$.

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Question 8

a. $w = 1 + yi, \quad y > 0$

$|w| = \sqrt{1+y^2}, \quad \text{Arg}(w) = \tan^{-1}(y), \quad w = \sqrt{1+y^2}\,\text{cis}\left(\tan^{-1}(y)\right)$ A1

$w^5 = \left(1+y^2\right)^{\frac{5}{2}}\text{cis}\left(5\tan^{-1}(y)\right) = \left(1+y^2\right)^{\frac{5}{2}}\left[\cos\left(5\tan^{-1}(y)\right) + i\sin\left(5\tan^{-1}(y)\right)\right]$

since $\text{Im}\left(w^5\right) = 0, \Rightarrow \sin\left(5\tan^{-1}(y)\right) = 0,$ but $\sin(n\pi) = 0, \quad n \in Z$

$5\tan^{-1}(y) = n\pi$

$\tan^{-1}(y) = \dfrac{n\pi}{5}$ let $n = 1, 2$ M1

$y = \tan\left(\dfrac{\pi}{5}\right), \ \tan\left(\dfrac{2\pi}{5}\right)$ as two answers,

but $y > 0$, tan is positive in the first and third quadrants

$y = \tan\left(\dfrac{\pi}{5}\right), \tan\left(\dfrac{2\pi}{5}\right), \tan\left(\dfrac{6\pi}{5}\right), \tan\left(\dfrac{7\pi}{5}\right), \ldots\ldots \tan\left(\dfrac{\pi}{5} + k\pi\right), \ \tan\left(\dfrac{2\pi}{5} + k\pi\right)$

$y = \tan\left(\dfrac{(5k+1)\pi}{5}\right), \tan\left(\dfrac{(5k+2)\pi}{5}\right), \quad k \in Z$ A1

b. $(1+x)^5 = 1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5$

$w^5 = (1+yi)^5 = 1 + 5yi + 10(yi)^2 + 10(yi)^3 + 5(yi)^4 + (yi)^5$

$= 1 + 5yi + 10y^2i^2 + 10y^3i^3 + 5y^4i^4 + y^5i^5$

Since $i^2 = -1, \ i^3 = -i, \ i^4 = 1, \ i^5 = i$

$w^5 = (1+yi)^5 = 1 + 5yi - 10y^2 - 10y^3i + 5y^4 + y^5i$

$= \left(1 - 10y^2 + 5y^4\right) + i\left(5y - 10y^3 + y^5\right)$ M1

$\text{Im}\left(w^5\right) = 0 \Rightarrow 5y - 10y^3 + y^5 = y\left(y^4 - 10y^2 + 5\right) = 0$

since $y > 0 \quad y^4 - 10y^2 + 5 = 0$

$y^2 = \dfrac{10 \pm \sqrt{100-20}}{2} = \dfrac{10 \pm \sqrt{80}}{2} = \dfrac{10 \pm 4\sqrt{5}}{2}$

$y^2 = 5 \pm 2\sqrt{5}$

$y = \sqrt{5 - 2\sqrt{5}} \ , \ \sqrt{5 + 2\sqrt{5}} \ , \ y > 0$ A1

since $\tan\left(\dfrac{2\pi}{5}\right) > \tan\left(\dfrac{\pi}{5}\right)$

$\tan\left(\dfrac{\pi}{5}\right) = \sqrt{5 - 2\sqrt{5}} \qquad a = 5, \ b = -2$ A1

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Question 9

a. $I_1 = \int_0^1 \frac{x}{(x+1)^2}\,dx$

let $u = x+1,\ \frac{du}{dx} = 1,\ \ x = u-1$

terminals when $x = 0,\ u = 1$ and when $x = 1,\ u = 2$

$$I_1 = \int_1^2 \left(\frac{u-1}{u^2}\right) du$$

$$I_1 = \int_1^2 \left(\frac{1}{u} - u^{-2}\right) du = \left[\log_e(|u|) + \frac{1}{u}\right]_1^2$$

$$I_1 = \left[\left(\log_e(2) + \frac{1}{2}\right) - \left(\log_e(1) + \frac{1}{1}\right)\right]$$

$$I_1 = \log_e(2) - \frac{1}{2}$$ A1

b. $I_n = \int_0^1 \frac{x^n}{(x+1)^2}\,dx = \int_0^1 x^n \times \frac{1}{(x+1)^2}\,dx$ using integration by parts

let $u = x^n \qquad \frac{dv}{dx} = \frac{1}{(x+1)^2}$

$$\frac{du}{dx} = nx^{n-1} \qquad v = \int \frac{1}{(x+1)^2}\,dx = \int (x+1)^{-2}\,dx = -(x+1)^{-1} = \frac{-1}{x+1}$$ M1

$$I_n = \left[\frac{-x^n}{x+1}\right]_0^1 + \int_0^1 \frac{nx^{n-1}}{(x+1)}\,dx$$

$$I_n = \left(\frac{-(1^n)}{2} + 0\right) + \int_0^1 \frac{nx^{n-1}(x+1)}{(x+1)^2}\,dx$$

$$I_n = -\frac{1}{2} + n\int_0^1 \frac{x^n + x^{n-1}}{(x+1)^2}\,dx$$ A1

$$I_n = -\frac{1}{2} + n\left[\int_0^1 \frac{x^n}{(x+1)^2}\,dx + \int_0^1 \frac{x^{n-1}}{(x+1)^2}\,dx\right]$$

$$I_n = -\frac{1}{2} + n(I_n + I_{n-1}) = -\frac{1}{2} + nI_n + nI_{n-1}$$ M1

$$I_n - nI_n = (1-n)I_n = -\frac{1}{2} + nI_{n-1}, \qquad I_n = -\frac{1}{2(1-n)} + \frac{n}{1-n}I_{n-1}$$

$$I_n = \frac{1}{2(n-1)} - \frac{n}{n-1}I_{n-1}$$

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c. $I_3 = \int_0^1 \frac{x^3}{(x+1)^2}\,dx$

using the reduction formula

let $n=2,\ \ I_2 = \frac{1}{2} - 2I_1 = \frac{1}{2} - 2\left(\log_e(2) - \frac{1}{2}\right) = \frac{3}{2} - 2\log_e(2)$ from **a.**

let $n=3,\ \ I_3 = \frac{1}{4} - \frac{3}{2}I_2 = \frac{1}{4} - \frac{3}{2}\left(\frac{3}{2} - 2\log_e(2)\right)$

$I_3 = 3\log_e(2) - 2$ A1

Question 10

a. $X(4,0,0),\ Y(0,-3,0),\ Z(0,0,6),\quad \overrightarrow{OX} = 4\underset{\sim}{i},\quad \overrightarrow{OY} = -3\underset{\sim}{j},\quad \overrightarrow{OZ} = 6\underset{\sim}{k}$

$\overrightarrow{XY} = \overrightarrow{OY} - \overrightarrow{OX} = -4\underset{\sim}{i} - 3\underset{\sim}{j},\quad \overrightarrow{XZ} = \overrightarrow{OZ} - \overrightarrow{OX} = -4\underset{\sim}{i} + 6\underset{\sim}{k},$

$$\overrightarrow{XY} \times \overrightarrow{XZ} = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ -4 & -3 & 0 \\ -4 & 0 & 6 \end{vmatrix} = -18\underset{\sim}{i} + 24\underset{\sim}{j} - 12\underset{\sim}{k} = -6\left(3\underset{\sim}{i} - 4\underset{\sim}{j} + 2\underset{\sim}{k}\right)$$

Let the normal to the plane be $\underset{\sim}{n} = 3\underset{\sim}{i} - 4\underset{\sim}{j} + 2\underset{\sim}{k}$

The equation of the plane is $3(x-4) - 4y + 2z = 0$

$3x - 4y + 2z = 12$ A1

b. line $\frac{x - x_1}{a} = y + 2 = \frac{z+1}{-1} = t$

$x = x_1 + ta,\quad y = -2 + t,\quad z = -1 - t$

substitute into the plane $3x - 4y + 2z = 12$

$3(x_1 + ta) - 4(-2 + t) + 2(-1 - t) = 12$

$3x_1 + 3ta + 8 - 4t - 2 - 2t = 12$

$(3a - 6)t = 6 - 3x_1$

i. intersect at a unique point $a \neq 2,\quad x_1 \in R.$ A1

ii. do not intersect $a = 2,\quad x_1 \neq 2$ A1

End of detailed answers for the
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